NETWORKING · NET-09

ACLs & Wildcard Masks

Use this when: you permit or deny traffic on a router or switch, or lock down who can SSH to it.

BreakFixLearn

CCNA 200-301 · topic 5.6
Sheet v1.0 · Oct 2026

Scan to visit

Wildcard masks and how an ACL decides

Wildcard = 255.255.255.255 minus the subnet mask

`0` bit = must match | `1` bit = ignore

One host	`192.0.2.10 0.0.0.0`	= `host 192.0.2.10`
/24	`192.0.2.0 0.0.0.255`	192.0.2.0 to .255
/26	`192.0.2.64 0.0.0.63`	192.0.2.64 to .127
Everything	`0.0.0.0 255.255.255.255`	= `any`

PER PACKET

1. Check lines top to bottom
2. First match wins; stop there
3. No match? Hidden **deny any** at the end

Standard vs extended

	STANDARD	EXTENDED
Numbers	1-99, 1300-1999	100-199, 2000-2699
Matches	Source IP only	Protocol, source, dest, ports
Place it	Near the destination	Near the source
Named	`ip access-list standard|extended NAME`	

Rules of thumb

One per	Interface, direction, protocol
in / out	In = before routing; out = after
Order	Specific lines above general ones
Ports	`eq` `gt` `lt` `neq` `range`

Extended, named

```
ip access-list extended WEB-IN
 remark HTTPS to the web server
 10 permit tcp any host 192.0.2.10 eq 443
 20 permit tcp 198.51.100.0 0.0.0.255 host 192.0.2.10 eq 22
 30 deny ip any any log
interface g0/1
 ip access-group WEB-IN in
```

Sequence numbers let you insert (`15 permit ...`) or delete (`no 20`).

Lock down SSH

```
access-list 10 permit host 198.51.100.5
line vty 0 15
 access-class 10 in
 transport input ssh
```

Only the admin PC can SSH in. On lines, use a numbered standard ACL with `access-class`, not `access-group`.

Check it

`show access-lists`	Lines and hit counts
`show ip interface g0/1`	Which ACL is applied, which way
`show run | section access-list`	The config
`clear access-list counters`	Reset hit counts
`deny ip any any log`	See what the hidden deny drops

Don't lock yourself out

DANGER A new ACL on your management path can cut you off. Permit your admin subnet first, and set `reload in 10` as a safety net.

CAUTION An ACL with only `deny` lines blocks everything.

CAUTION An outbound ACL does not filter traffic the router itself sends.

REMEMBER

- /27 mask .224 = wildcard **.31**
- First match wins
- Hidden **deny any** at the end

Sources: Cisco Catalyst 9300 IPv4 ACL guide (IOS XE 17.9) · Cisco IOS XE ACL configuration guide · CCNA 200-301
CC BY-NC 4.0 · NET-09 v1.0